

MST – 1 (Sep 2024)
BASIC ELECTRONICS ENGINEERING (ECEB1123T)
1st YEAR – SEMESTER - 1
PUNJABI UNIVERSITY, PATIALA

MARKS: 15

TIME: 1 HOUR

SECTION – A (5 Marks) (Attempt all questions)

- Q1. (a) Draw a circuit diagram and waveform of the shunt negative clipper with negative bias? (1)
(b) Draw the schematic symbol of UJT and an NPN transistor? (1)
(c) Why NPN is preferred over PNP? (1)
(d) Derive the relation between α and β ? (1)
(e) Define cut-in voltage? (1)

SECTION – B (10 Marks) (Attempt any 2 Questions)

- Q2. (a) Write down the differences between BJT and FET? (2)
(b) What is a zener diode. Explain its use as a voltage regulator? (3)
- Q3. Explain the construction of N-channel JFET and elaborate its operation with the help of drain/output characteristics and transfer/input characteristics? (5)
- Q4. Explain the Half-wave rectifier with the help of a circuit diagram and output waveform. Also, derive its efficiency? (5)